



On the performance of conically shaped columns: Theory and practice

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ABSTRACT

The chromatographic performance (speed, efficiency, and gradient peak capacity for the same analysis time) of conical columns are investigated from fundamental and experimental viewpoints. A stainless steel, conically shaped column (2.1 mm i.d./4.2 mm i.d. × 15 cm long, 0.4° opening angle) was prepared in-house and packed with 5 μm XBridge-C₁₈ fully porous particles. Its performance was compared to that of a conventional 3.0 mm × 15 cm cylindrical column packed with the same batch of particles. Both Giddings' theory of non-uniform columns and experiments agree and show that, irrespective of flow direction, the conical column is 15% less efficient than the conventional column. Remarkably, Blumberg's theory of band broadening in gradient elution mode predicts that conical columns may outperform conventional cylindrical columns if the ratio of their outlet i.d. to their inlet i.d. is 0.95 and 0.80 for small molecule and peptide mixtures, respectively. The maximum relative gain is marginal as it does not exceed a few percents. The theory reveals that the flow direction should be from the wide to the narrow end of the conical column in order to deliver the highest peak capacity. In agreement with the theory, the observed losses in absolute peak capacity for the same analysis time are 14.5% (narrow to wide end) and only 11.0% (wide to the narrow end) for small molecules (*n*-alkanophenones). They are 14.2% (narrow to wide end) and only 8.5% (wide to the narrow end) for peptide samples (bombesin). Additionally, conical columns reduce peak tailing with respect to standard columns. They are suitable column technology for ultra-fast gradient separations as they also minimize sample dispersion through the narrow i.d. outlet frit.

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1. Introduction

Chromatographic columns are almost exclusively cylindrical open tubes, irrespective of whether they are open, packed, coated, or surface-bonded columns for applications in gas, liquid, and supercritical fluid chromatography. The reason is that tubes with non-uniform cross-section area are inherently unfavorable under isocratic condition as they induce changes in the average linear velocity along the tube [1]. The conical column is a non-uniform column in the sense of Giddings' theoretical treatment of band broadening along chromatographic columns [2]. Its efficiency is inevitably lower than that of the cylindrical column. Additionally, from a practical and economic viewpoint, it is much more convenient and cheaper to prepare metal, glass, or plastic tubes with uniform than with variable cross-section area along their length. For these reasons, the research interests in conical columns have been very limited. Some authors have shown that conical columns present some practical advantages for certain applica-

tions by reducing the impact of non-idealities in conventional columns. For instance, conical monolithic columns have been prepared to eliminate wall channeling and maximize loading/recovery of deoxyribonucleic acids [3,4]. Also, conical preparative chromatographic columns with 7 to 15° angle were prepared to potentially compensate for the distorted flow profile across the large diameter of preparative columns [5–7]. However, to the best of our knowledge, no research effort have been devoted to the preparation and to the study of stainless steel, analytical size, conical columns.

The basic fundamentals of isocratic and gradient liquid chromatography (GLC) have been established since the early ages of liquid chromatography and later refined for some specific cases [1,8–21]. They are almost exclusively based on cylindrical geometry for the column tube. Very few theoretical treatments have been reported regarding the loadability and performance of conical columns under isocratic conditions [22–24]. In particular, Said [22] demonstrated the advantage of conical columns over composite columns (having the same length and the same volume) in terms of resolution power in isocratic conditions. Yet, the fundamental treatment of band broadening along conical columns is still missing in gradient elution mode.

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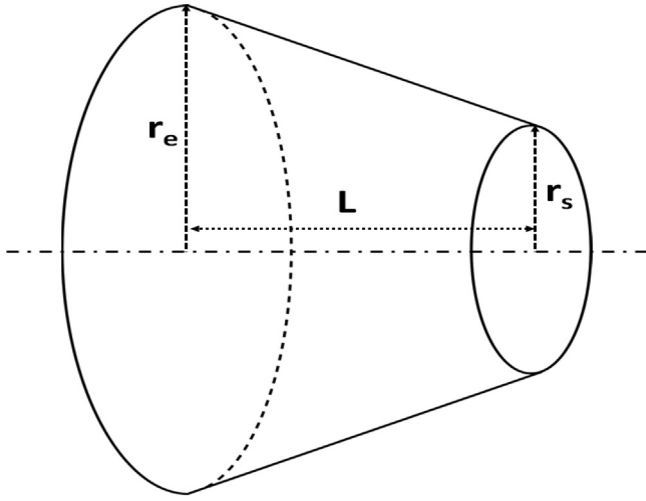


Fig. 1. Representation of a conical-shaped column by a truncated cone of height L and radii r_e and r_s . By convention, the direction of the flow is from left to right.

In this work, the fundamental concepts of band broadening in isocratic and gradient elution modes [1,13,14,21] are first extended to conical-shaped columns. The local specific permeability, K_0 , of conical and cylindrical columns is assumed to remain uniform along their length while the local linear chromatographic velocity and the local plate height are allowed to vary [14]. In the theory developed in this work, the non-zero radial component of the local velocity vector was not considered, so, the radial pressure drop across the column was neglected. Since the radial component of the velocity vector only affects transverse dispersion, it has no impact on the measured retention and axial plate height of the conical column. However, the non-zero radial component of the local velocity vector is accounted in this work in the sense that the axial velocity is continuously changing along the column. In the first part of this work, the calculations assume that the plate height is uniform along the column in order to assess the potential of conical columns in gradient elution. In the second part and based on experimental data, the model also accounts for the plate height variation along the column so that experimental and theoretical peak capacities can be directly compared. The intrinsic impact of the column geometry on the chromatographic speed, apparent plate height, and gradient peak capacity is investigated at constant pressure and constant analysis time. The results are reported as a function of the ratio, ρ_s , of the outlet i.d. to the inlet i.d. of the conical column. Secondly, the theoretical predictions are tested experimentally based on the fabrication of a prototype conical column (2.1 mm i.d./4.2 mm i.d. $\times$ 15 cm long, 0.4° opening angle) packed with 5 μ m XBridge-C18 fully porous particles. The experimental column efficiency and gradient peak capacity of the conical column are compared to those of a conventional 3.0 mm $\times$ 15 cm cylindrical column packed with the same batch of 5 μ m particles. The agreement between theory and experiments is discussed in depth and the potential applications of conical columns is proposed.

2. Theory

2.1. Definitions

The conically shaped chromatographic column is assumed to be a truncated cone as that shown in Fig. 1. The inlet and outlet i.d. of the conical column are noted r_e and r_s , respectively. The ratio ρ_s is then defined by:

$$\rho_s = \frac{r_s}{r_e} \quad (1)$$

The height of the truncated cone is the column length, L . The origin for the axial coordinate ($z=0$) is taken at the column entrance. The dimensionless axial position, ζ , is then defined by:

$$\zeta = \frac{z}{L} \quad (2)$$

The dimensionless inner radius is defined by:

$$\rho = \frac{r}{r_e} \quad (3)$$

Accordingly, ρ is linearly changing with increasing the dimensionless position ζ :

$$\rho(\zeta) = 1 + (\rho_s - 1)\zeta \quad (4)$$

The average chromatographic linear velocity, $u_0(\zeta)$, at the dimensionless position ζ is then given by:

$$u_0(\zeta) = \frac{u_0(0)}{\rho(\zeta)^2} \quad (5)$$

where $u_0(0)$ is the average chromatographic linear velocity at the column entrance ($\zeta=0$, $\rho=1$).

2.2. Pressure drop along conical columns

The local permeability equation along a column slice in between the dimensionless axial positions ζ and $\zeta + d\zeta$ is written:

$$dP = -\frac{\eta}{K_0} \epsilon_t u_0(\zeta) L d\zeta \quad (6)$$

$$= -\frac{\eta \epsilon_t L}{K_0} u_0(0) \frac{d\zeta}{(1 + [\rho_s - 1]\zeta)^2} \quad (7)$$

where dP is the elementary pressure drop ($dP < 0$ for $d\zeta > 0$) along the column slice, η is the dynamic viscosity of the eluent, K_0 is the specific permeability of the chromatographic bed assumed to be uniform along the column, and ϵ_t is the total porosity of the column.

Integration of Eq. (7) from the column inlet ($\zeta=0$) to the column outlet ($\zeta=1$) gives:

$$P_e - P_s = \frac{\eta \epsilon_t L}{K_0} u_0(0) \int_0^1 \frac{d\zeta}{(1 + [\rho_s - 1]\zeta)^2} \quad (8)$$

$$= \frac{\eta \epsilon_t L}{K_0} u_0(0) \frac{1}{1 - \rho_s} \left[\frac{1}{\rho_s} - 1 \right] \quad (9)$$

$$= \frac{\eta L}{K_0} \frac{F_v}{\pi \rho_s r_e^2} \quad (10)$$

where F_v is the applied flow rate.

Eq. (10) shows that a conical column (r_e , r_s) has the same permeability as that of a cylindrical column having the same length and an inner radius $r_{eq} = \sqrt{\rho_s r_e^2} = \sqrt{r_e r_s}$ (geometric mean). For a 2.1 mm i.d./4.2 mm i.d. truncated cone, $2r_{eq} = 2.97$ mm.

2.3. Isocratic elution

2.3.1. Elution time

Let us define the increment of the dimensionless hold-up time $d\tau_m(\zeta)$ as:

$$d\tau_m(\zeta) = \frac{dt_m(\zeta)}{\frac{L}{u_0(0)}} \quad (11)$$

$$= \frac{\frac{L d\zeta}{u_0(\zeta)}}{\frac{L}{u_0(0)}} \quad (12)$$

$$= \rho(\zeta)^2 d\zeta \quad (13)$$

where $t_m(\zeta)$ is the hold-up time at the dimensionless axial position ζ . According to Eq. (11), the elution times are all normalized to the reference time $\frac{L}{u_0(0)}$.

Integration of Eq. (11) gives the expression of the dimensionless hold-up time as a function of the dimensionless axial position ζ :

$$\tau_m(\zeta) = \int_0^\zeta [1 + (\rho_s - 1)u]^2 du \quad (14)$$

$$= \frac{1}{3(\rho_s - 1)} ([1 + (\rho_s - 1)\zeta]^3 - 1) \quad (15)$$

$$= \zeta \left[1 + (\rho_s - 1)\zeta + \frac{(\rho_s - 1)^2}{3} \zeta^2 \right] \quad (16)$$

where u is a dummy integration variable.

Under isocratic elution, the retention factor k is constant during the migration of the analyte band. The dimensionless retention time, $\tau_e(1)$, is given at the column outlet for $\zeta = 1$. Accordingly,

$$\tau_e(1) = (1 + k)\tau_m(1) \quad (17)$$

$$= \frac{1}{3}(1 + k)(1 + \rho_s + \rho_s^2) \quad (18)$$

Eq. (18) reveals that a conical column (r_e , r_s) has the same hold-up volume as that of a cylindrical column, which has the same length, the same total porosity, and an inner radius equal to $r_{eq} = \sqrt{\frac{r_e^2 + r_e r_s + r_s^2}{3}}$. For a 2.1 mm i.d./4.2 mm i.d. truncated cone, $2r_{eq} = 3.21$ mm.

2.3.2. Temporal variance

The elementary increment of the dimensionless temporal variance, $d\sigma_t^2(\zeta)$, is defined by:

$$d\sigma_t^2(\zeta) = \frac{\frac{H(\zeta)Ld\zeta}{u_0(\zeta)^2}}{\left(\frac{L}{u_0(0)}\right)^2} \quad (19)$$

$$= \frac{H(\zeta)}{L} \rho(\zeta)^4 d\zeta \quad (20)$$

where $H(\zeta)$ is the local plate height at the dimensionless axial position ζ . According to Eq. (19), the temporal variance is normalized to the reference time squared $\left(\frac{L}{u_0(0)}\right)^2$.

If the local plate height is assumed to be uniform along the whole column length ($H(\zeta) = H$), the integration of Eq. (20) gives the expression of the dimensionless temporal variance:

$$\sigma_t^2(\zeta) = \frac{H}{L} \int_0^\zeta [1 + (\rho_s - 1)u]^4 du \quad (21)$$

$$= \frac{H}{L} \frac{1}{5(\rho_s - 1)} ([1 + (\rho_s - 1)\zeta]^5 - 1) \quad (22)$$

$$= \zeta \left[1 + 2(\rho_s - 1)\zeta + 2(\rho_s - 1)^2 \zeta^2 + (\rho_s - 1)^3 \zeta^3 + \frac{(\rho_s - 1)^4}{5} \zeta^4 \right] \frac{H}{L} \quad (23)$$

The dimensionless temporal variance expected at the column outlet is given for $\zeta = 1$. Additionally, for a retained compound, Eq. (23) gives the following expression for σ_t^2 :

$$\sigma_t^2(1) = \frac{1}{5}(1 + k)^2 (1 + \rho_s + \rho_s^2 + \rho_s^3 + \rho_s^4) \frac{H}{L} \quad (24)$$

2.3.3. Apparent column plate height

Under isocratic conditions, Giddings defined and calculated the apparent plate height, $\bar{H}$, from the first and second central moment of the concentration distribution in the time domain [1]. From Eqs. (18) and (24), $\bar{H}$ is written:

$$\bar{H} = L \frac{\sigma_t^2(1)}{\tau_e^2(1)} \quad (25)$$

$$= \frac{9}{5} \frac{1 + \rho_s + \rho_s^2 + \rho_s^3 + \rho_s^4}{(1 + \rho_s + \rho_s^2)^2} H \quad (26)$$

Note that the above expression is invariant after the transformation $\rho_s \rightarrow \frac{1}{\rho_s}$, e.g., the result does not depend on the flow direction along the conical column. As expected, $\bar{H} = H$ for cylindrical columns and $\bar{H} = 1.8H$ for a cone ($\rho_s \rightarrow \infty$ or 0). So, the maximum relative loss in efficiency is 80%. For a 2.1 mm i.d./4.2 mm i.d. truncated cone, $\bar{H} = 1.14H$. Its efficiency is then expected to be only 14% smaller than that of cylindrical column. The corresponding relative plate height, h_r , is defined by:

$$h_r = \frac{\bar{H}}{H} \geq 1 \quad (27)$$

2.4. Gradient elution

In this section, it is assumed that the solvent gradient is linear and not distorted upon migration along the chromatographic column. Additionally, the variation of the retention factor, $k(\varphi)$, of the analytes as a function of the volume fraction, φ , of the strong solvent in the eluent follows the well-known Linear Solvent Strength Model (LSSM) [8]. Accordingly,

$$k(\varphi) = k_0 \exp(-S[\varphi - \varphi_0]) \quad (28)$$

where φ_0 is the starting volume fraction of the strong solvent, S and k_0 are the two LSSM retention parameters.

The volume fraction of the strong eluent in the eluent can be expressed for any dimensionless time τ and axial position ζ [9]:

$$\varphi(\tau, \zeta) = \varphi_0 + \frac{\beta L}{u_0(0)} [\tau - \tau_m(\zeta)] \quad (29)$$

$$= \varphi_0 + \frac{\beta L}{u_0(0)} \left(\tau - \zeta \left[1 + (\rho_s - 1)\zeta + \frac{(\rho_s - 1)^2}{3} \zeta^2 \right] \right) \quad (30)$$

where β is the temporal gradient steepness and the general expression of the dimensionless hold-up time $\tau_m(\zeta)$ at axial position ζ is given by Eq. (16).

2.4.1. Elution time

The general gradient equation and the dimensionless gradient retention time, $\tau_e(\zeta)$, is written [8,10] at any position ζ along the column:

$$\int_0^{\tau_e(\zeta) - \tau_m(\zeta)} \frac{d\tau_s}{k_0 \exp\left(-S\beta \frac{L}{u_0(0)} \tau_s\right)} = \tau_m(\zeta) \quad (31)$$

where $\tau_s = \tau_e - \tau_m$ is the dimensionless residence time in the stationary phase.

Let us define the intrinsic gradient steepness G as:

$$G = S\beta \frac{L}{u_0(0)} \quad (32)$$

Integration of Eq. (31) from $\tau_s = 0$ to $\tau_e(\zeta) - \tau_m(\zeta)$ (left-hand-side of Eq. (31)) and from $\tau_m = 0$ to $\tau_m(\zeta)$ (right-hand-side of Eq. (31)) gives the gradient elution time, $\tau_e(\zeta)$, at any axial position column

ζ along the conical column:

$$\tau_e(\zeta) = \zeta \left[1 - (1 - \rho_s)\zeta + \frac{(1 - \rho_s)^2}{3}\zeta^2 \right] + \frac{1}{G} \ln \left(1 + Gk_0\zeta \left[1 - (1 - \rho_s)\zeta + \frac{(1 - \rho_s)^2}{3}\zeta^2 \right] \right) \quad (33)$$

In particular, for $\zeta = 1$, the gradient elution time at the column outlet, $\tau_e(1)$, is written:

$$\tau_e(1) = \frac{1}{3} [1 + \rho_s + \rho_s^2] + \frac{1}{G} \ln \left(1 + G \frac{k_0}{3} [1 + \rho_s + \rho_s^2] \right) \quad (34)$$

An important parameter in gradient elution is the local retention factor during elution $k_e(\zeta)$. By first substituting the expression of $\tau_e(\zeta)$ (Eq. (33)) into Eq. (30) (which gives the volume fraction, $\varphi(\tau_e(\zeta), \zeta)$, of the strong solvent at elution) and then introducing the expression of $\varphi(\tau_e(\zeta), \zeta)$ into Eq. (28), $k_e(\zeta)$ is written:

$$k_e(\zeta) = \frac{k_0}{1 + Gk_0\zeta \left[1 + (\rho_s - 1)\zeta + \frac{(\rho_s - 1)^2}{3}\zeta^2 \right]} \quad (35)$$

In particular, at the column outlet or for $\zeta = 1$:

$$k_e(1) = \frac{k_0}{1 + G \frac{k_0}{3} [1 + \rho_s + \rho_s^2]} \quad (36)$$

2.4.2. Temporal variance

The variation of the spatial variance, σ_z^2 , with increasing the position of the band center obeys the following general differential equation derived by Blumberg for smooth media [13,14]. In smooth media, the plate height, $H(z)$, may depend on the axial position z along the column but it is assumed to remain uniform across the peak width. This differential equation is written [14]:

$$\frac{d\sigma_z^2}{dz} = H(z) + 2 \frac{\sigma_z^2}{u_e} \left[\frac{\partial u_e}{\partial x} \right]_{x=0} \quad (37)$$

where $H(z)$ is the local zonal plate height at position z and $u_e(z)$ is the local linear migration velocity of the analyte when the band center reaches the location z ,

$$u_e(z) = \frac{u_0(z)}{1 + k_e(z)} \quad (38)$$

In Eq. (37), the variable distance x accounts for the spatial departure from the fixed position z of the band center. The partial derivative means that the derivative should be taken at a fixed elution time $t = t(z)$ when the center of mass has reached the position z along the column. The dimensionless local linear migration velocity is defined as:

$$v_e = \frac{u_e}{u_0(0)} \quad (39)$$

v_e is then expressed as a function of the dimensionless position ζ along the conical column from Eqs. (4, 5), and (35):

$$v_e(\zeta) = \frac{1}{\rho(\zeta)^2} \frac{1}{1 + k_e(\zeta)} \quad (40)$$

Note that $v_e(\zeta)$ is affected by both the column geometry through $\rho(\zeta)$ and the retention factor at position ζ , $k_e(\zeta)$.

Assuming that the spatial peak width is much smaller than the column length, then, the partial derivative of $v_e(\xi, \zeta)$ with respect to $\xi = \frac{x}{L}$ ($x \ll L$) at constant ζ in Eq. (37) can be estimated from the first order development of $\rho(\zeta + \xi)$ and $k_e(\zeta + \xi)$ around ζ . At a fixed

time or at a fixed position ζ of the band center, it can be shown after elementary algebra (chain rule) that:

$$\left[\frac{\partial \frac{1}{\rho(\zeta + \xi)^2}}{\partial \xi} \right]_{\xi=0} = - \frac{2}{\rho(\zeta)^3} (\rho_s - 1) \quad (41)$$

and

$$\left[\frac{\partial k_e(\zeta + \xi)}{\partial \xi} \right]_{\xi=0} = Gk_e(\zeta) [1 + (\rho_s - 1)\zeta]^2 > 0 \quad (42)$$

Therefore

$$\frac{2}{v_e} \left[\frac{\partial v_e}{\partial \xi} \right]_{\xi=0} = -2 \left(G \frac{k_e(\zeta)}{1 + k_e(\zeta)} [1 + (\rho_s - 1)\zeta]^2 + \frac{2[\rho_s - 1]}{\rho(\zeta)} \right) \quad (43)$$

Accordingly, the dimensionless differential equation to solve in the dimensionless spatial variance $y = \frac{\sigma_z^2}{L^2}$ is then

$$\frac{dy}{d\zeta} + f(\zeta)y = \frac{H(z)}{L} \quad (44)$$

where $f(\zeta)$ is an explicit function of the variable ζ given by:

$$f(\zeta) = 2 \left(G \frac{k_e(\zeta)}{1 + k_e(\zeta)} [1 + (\rho_s - 1)\zeta]^2 + \frac{2[\rho_s - 1]}{\rho(\zeta)} \right) \quad (45)$$

The general solution $y(\zeta)$ of this nonhomogeneous, linear, first-order differential equation (44) is directly given from [21]:

$$y(\zeta) = \frac{1}{L} e^{-\int_0^\zeta f(u)du} \left[\int_0^\zeta e^{\int_0^u f(v)dv} H(u)du \right] \quad \text{with } y(0) = 0 \quad (46)$$

where u and v are dummy variables. The solution at the column outlet is found for $\zeta = 1$. The three integrals in Eq. (46) can be numerically calculated using Excel software.

The dimensionless temporal peak variance expected under gradient elution for a conical column at its outlet ($\zeta = 1$) is then finally obtained from the following expression:

$$\sigma_{t,g}^2 = \frac{y(1)}{v_e(1)^2} = y(1)\rho_s^4 [1 + k_e(1)]^2 \quad (47)$$

It is important to recall that this model of gradient chromatography elaborated for conical columns assumes that the column media is smooth ($H(z, x) = H(z)$).

2.4.3. Peak capacity for the same analysis time

Let us consider N different peaks eluted with dimensionless standard deviations $\sigma_{t,i}$ ($i = 1$ to N). The peaks are spread uniformly between the dimensionless gradient elution times $\tau_{e,1}$ ($i = 1$) and $\tau_{e,N}$ ($i = N$). The 4σ peak capacity, P_c , can be calculated from the average peak width of the N peaks:

$$P_c = 1 + \frac{\tau_{e,N} - \tau_{e,1}}{\frac{1}{N} \sum_{i=1}^N 4\sigma_{t,i}} \quad (48)$$

The gradient separation time (in seconds or minutes), t_g , is given from the definition of the dimensionless times (see Eq. (13)):

$$t_g = \frac{L}{u_0(0)} (\tau_{e,N} - \tau_{e,1}) \quad (49)$$

The peak capacity for the same analysis time, $P_{c,t}$, is defined as the maximum of peaks that can be separated per unit time. It is then written:

$$P_{c,t} = \frac{P_c}{t_g} \quad (50)$$

$$= \frac{u_0(0)}{L} \left[\frac{1}{\tau_{e,N} - \tau_{e,1}} + \frac{N}{4} \frac{1}{\sum_{i=1}^N \sigma_{t,i}} \right] \quad (51)$$

3. Experimental

3.1. Chemicals

The mobile phases used are mixtures of acetonitrile and water. Both solvents were Optima grade from Fisher Scientific (Fair Lawn, NJ, USA). 1 mL ampules of trifluoro-acetic acid (TFA) from Sigma-Aldrich (Suwannee, GA, USA) were used for the gradient elution of the peptide bombesin. The small molecules (acetophenone, propiophenone, butyrophenone, valerophenone, and hexanophenone) and bombesin were all purchased from Sigma-Aldrich (Suwannee, GA, USA).

3.2. Instrument

The i-class Acquity UHPLC liquid chromatograph (Waters, Milford, MA, USA) was used for all the chromatographic measurements. This instrument includes a binary (pump heads A and B) solvent delivery system, an auto-sampler equipped with a 15 μ L flow-through-needle loop connected from the injection port to the injection valve by a 30 cm $\times$ 100 μ m tube (generating a pre-valve sample dispersion of 0.5 μ L²), a column manager (accommodating two columns and equipped with one inlet and one outlet switch valves), an active column preheater (31 cm $\times$ 75 μ m, room temperature to 90°C) and a photo diode array (PDA) detector equipped with a 125 nL flow cell. The whole system is operated by the Empower 3 Chromatography Data Software (Waters, Milford, MA, USA). The flow rate may range from 0 to 2 mL/min. The column manager temperature was set at 27 °C.

3.3. Columns

The conventional (control) column is a cylindrical 3.0 mm $\times$ 150 mm stainless steel column packed at constant flow rate with 5.0 μ m XBridge-C₁₈ particles (Waters, Milford, USA). The empty column volume is 1.06 cm³. The stainless steel conical column hardware is 150 mm long with inner diameters of 2.1 and 4.2 mm at both ends. The volume of the empty conical column is 1.21 cm³. This special conical-shaped tube was home-made according to proprietary techniques which cannot be released. It was packed at constant flow rate with the same 5.0 μ m XBridge-C₁₈ particles as those used to pack the 3.0 mm i.d. cylindrical column. The columns were sealed at their ends with 2.1 mm (conical column), 3.0 mm (cylindrical column), and 4.2 mm (conical column) i.d. frits (1 mm thick, 1 μ m porosity, 35% void fraction).

3.4. Chromatographic conditions

3.4.1. Plate height variation along the conical column

The variation of the plate height, $H(z)$, of acetophenone, propiophenone, butyrophenone, valerophenone, and hexanophenone as a function of the axial position z along the 2.1 mm/4.2 mm i.d. conical column was determined from the variation of the plate height, $H(F_v)$ of the 3.0 mm i.d. cylindrical column at a series of flow rates increasing from 0.204 to 0.306, 0.408, 0.510, 0.612, 0.714, and to 0.816 mL/min using a mixture of acetonitrile and water (75/25, v/v) at 27 °C. These flow rates recreate exactly the linear velocities at the axial positions $z=0, 2.5, 5, 7.5, 10, 12.5$, and 15 cm along the conical column when the flow direction is from the wide end (4.2 mm i.d.) to the narrow end (2.1 mm i.d.) and the flow rate is set at 0.4 mL/min. They recreate exactly the linear velocities at the axial positions $z=15, 12.5, 10, 7.5, 5, 2.5$, and 0 cm along the conical column when the flow direction is from the narrow end (2.1 mm i.d.) to the wide end (4.2 mm i.d.) at the same flow rate of 0.4 mL/min.

3.4.2. Small molecules: alkanophenones

The five alkanophenones were eluted with a mixture of acetonitrile and water (75/25, v/v) under isocratic conditions at 27 °C. The sample mixture was prepared by dissolving 5 μ L of each alkanophenone in 1.5 mL of acetonitrile/water mixture (60/40, v/v). The injection volume is 0.5 μ L. The gradient time is 5 min. The starting and final volume fractions of acetonitrile in water are 60 and 95%, respectively. The flow rate is set at 0.35 and 0.40 mL/min for the cylindrical and conical columns, respectively. The detection wavelength was set at 265 nm for a sampling rate of 10 Hz.

3.4.3. Peptide: bombesin

The concentration of the peptide bombesin (14 amino acids: Pyr-Gln-Arg-Leu-Gly-Asn-Gln-Trp-Ala-Val-Gly-His-Leu-Met-NH₂, MW=1619.85 g/mol) is 0.1 g/L in a water-rich mixture (95% in volume) of acetonitrile and water containing 0.1% TFA. It is eluted under gradient conditions at 27 °C. The injection volume is 3.0 μ L. The gradient time is 5 min. The starting and final volume fractions of acetonitrile in water are 10 and 55%, respectively. The flow rate is set at 0.35 and 0.40 mL/min for the cylindrical and conical columns, respectively. The detection wavelength was set at 278 nm for a sampling rate of 10 Hz.

4. Results and discussion

In the first theoretical part of this section, the reference cylindrical column selected for comparison purposes is such that it generates either the same pressure drop or the same analysis time as those of the conically shaped column. The parameter, ρ_s , of the conical column is varied in a range from 0.1 to 10. In the second experimental part, a home-made 2.1 mm/4.2 mm i.d. $\times$ 150 mm conical column ($\rho_s = 0.5$ or 2 depending on flow direction) is used and its performance is compared to that of a control 3.0 mm i.d. $\times$ 150 mm cylindrical column at constant speed. The theoretical predictions are then tested experimentally.

4.1. Theoretical predictions: conical/cylindrical columns

4.1.1. Separation impedance

According to Eq. (10), the pressure drop along the conical column at any given flow rate F_v is exactly the same as that of a cylindrical column having the same length L and an inner radius, r_{eq} , equal to the geometric mean of the inlet (r_e) and outlet (r_s) inner radii of the conical column. In the next investigations, the performances (elution times and peak widths) of the conical-shaped column (characterized by a fixed inner radius r_e and the variable ratio ρ_s) will be compared to those of a reference cylindrical column having the inner radius $r_{eq} = \sqrt{r_e r_s}$ and the same length L . All columns are packed identically with the same particles of size d_p and have the same external porosity ϵ_e . For the sake of comparison and in order to assess the intrinsic importance of the column shape (conical versus cylindrical) on the chromatographic properties, the local plate height, H , is assumed to be independent of the axial position for both columns. Therefore, for any fixed linear velocity $u_0(0) = \frac{F_v}{\epsilon_t \pi r_e^2}$, the pressure drop ΔP , the eluent viscosity η , the local plate height H , and the column length L are the same for both the conical and cylindrical columns. The classical separation impedance E is written [25]:

$$E(\rho_s) = \frac{t_m(\rho_s)\Delta P}{\eta N(\rho_s)^2} \quad (52)$$

where $t_m(\rho_s)$ is the hold-up time and $N(\rho_s) = \frac{L}{H}$ is the column efficiency.

One can then define a relative separation impedance e as:

$$e = E(\rho_s) \frac{\eta u_0(0)L}{H^2 \Delta P} = h_r(\rho_s)^2 \tau_m(\rho_s) \quad (53)$$

Note that $e = 1$ for conventional cylindrical columns.

4.1.2. Isocratic elution

In this section, the parameter ρ_s is covering the interval $[0.1; 10]$. The reference dimensionless hold-up time (τ_m), relative plate height (h_r), and impedance ($e = h_r^2 \tau_m$) of the cylindrical column are shown as solid brown lines in **Figure 1 in the Supplementary Material** as a function of the variable ρ_s . The same dimensionless chromatographic properties are represented as solid blue lines for the conical column. The three graphs in **Figure 1 in the Supplementary Material** reveal that, under isocratic conditions and at a fixed pressure drop, conical columns are not intrinsically advantageous over classical ones. Indeed, irrespective of the radius ratio ρ_s , the expected analysis time and column efficiency of conical columns are always larger than those of cylindrical columns. Therefore, their impedance is also larger. These results are in line with the general theories of non-uniform or composite columns [1,22]. For that all, questions remain 1) whether or not this loss in column performance is always significant and 2) for which interval of ρ_s values around unity the conical shape of a column does not affect too severely the performance of the equivalent cylindrical column.

It is useful to point out that such losses in column performance are not too excessive provided that the parameter ρ_s remains in the range $0.5 < \rho_s < 2$. Indeed, **Figure 2 in the Supplementary Material** shows that, when the inlet and the outlet column radii of a conical column differ by less than a factor 2, the hold-up time increases by less than 17%, the apparent plate height increases by less than 14%, and the separation impedance grows by less than 50%.

4.1.3. Gradient elution

In 1981, Poppe derived a differential equation for the variation of the spatial variance, σ_z^2 , of the sample peak with increasing the position z of its mass center as it progresses along classical cylindrical columns, e.g., when the linear velocity and the local plate height remain constant along the column length [12]. Accordingly,

$$\frac{dy}{d\zeta} = \frac{H}{L} - 2 \frac{k_e}{1+k_e} ay \quad (54)$$

where $y = \frac{\sigma_z^2}{L^2}$ is the dimensionless spatial variance and the coefficient a is defined by [12]:

$$a = \frac{1}{k_e} \left[\frac{\partial k_e}{\partial \xi} \right]_{\xi=0} \quad (55)$$

For a LSSM retention model, $\left[\frac{\partial k_e}{\partial \xi} \right]_{\xi=0} = Gk_e$, and Poppe derived this following equation [12]:

$$\frac{dy}{d\zeta} + 2G \frac{k_e}{1+k_e} y = \frac{H}{L} \quad (56)$$

The correctness of the general differential equation (44) derived in this work is immediately verified for $\rho_s = 1$ (cylindrical columns) and $H(z) = H$ (uniform plate height profile along the column). It is straightforward to show that the expression of the function $f(\zeta)$ in Eqs. (56) and (44) are strictly identical for $\rho_s = 1$. For $\rho_s \neq 1$ (conical columns), the analysis of the different terms in Eq. (45) reveals three contributions, which simultaneously control the intensity of either compression or expansion of the spatial peak width with respect to the spatial peak width expected under isocratic conditions. If the sum of this three terms is positive, the band is compressed. If it is negative, the band expands during gradient elution:

1. The first term accounts for the compression of the band caused by the increase of the retention factor from the rear to the front ends of the peak. This term is the same as the one derived by Poppe and it is always positive:

$$2G \frac{k_e(\zeta)}{1+k_e(\zeta)} > 0 \quad (57)$$

2. A second term accounts for the either compression ($\rho_s > 1$) or expansion ($\rho_s < 1$) of the band due to the decrease or increase, respectively, of the linear velocity along the column. This term is independent of the retention behavior and is purely of geometrical origin. It is equal to zero for cylindrical columns ($\rho_s = 1$):

$$\frac{4[\rho_s - 1]}{\rho(\zeta)} > 0 \quad \text{or} < 0 \quad (58)$$

It is positive for $\rho_s > 1$ (flow direction from the narrow to the wide column i.d.) and negative for $\rho_s < 1$ (flow direction from the wide to the narrow column i.d.).

Finally, the third term accounts for the coupling between the change in the retention factor and that of the migration linear velocity. This term can be either positive ($\rho_s > 1$) or negative ($\rho_s < 1$) and it is equal to zero for cylindrical columns ($\rho_s = 1$):

$$2G \frac{k_e(\zeta)}{1+k_e(\zeta)} (\rho_s - 1) \zeta [(\rho_s - 1) \zeta + 2] > 0 \quad \text{or} < 0 \quad (59)$$

It is noteworthy that this term is always positive (band compression) irrespective of the position ζ along the column when $\rho_s > 1$. If $\rho_s < 1$, this term is always negative for all ζ .

Accordingly, the sign of the sum of those three terms is always positive for $\rho_s > 1$ and the band is compressed in the spatial domain. However, at the same time, the linear velocity at the column outlet is the smallest (it is inversely proportional to ρ_s^2) and the trade-off between band compression and low migration linear velocity at the column outlet should be investigated on the basis of Eq. (47).

In contrast, the band expands in the space domain when the following quantity becomes negative:

$$\rho_s^3 + b\rho_s - b < 0 \quad (60)$$

where b is a positive quantity given by

$$b = 2 \frac{1+k_e(1)}{Gk_e(1)} > 0 \quad (61)$$

The critical value $\tilde{\rho}_s < 1$ for which this quantity becomes negative depends on the intrinsic gradient steepness at the column entrance (G) and on the retention factor of the analyte at elution ($k_e(1)$). $\tilde{\rho}_s$ is given by the unique positive solution of Eq. (1):

$$\tilde{\rho}_s = \sqrt[3]{b \left(\frac{1}{2} + \sqrt{\frac{1}{4} + \frac{1}{27}b} \right)} + \sqrt[3]{b \left(\frac{1}{2} - \sqrt{\frac{1}{4} + \frac{1}{27}b} \right)} \quad (62)$$

Similarly, a trade-off is expected for $\rho_s < \tilde{\rho}_s < 1$ as the elution speed is maximum at the column outlet.

The main goals of the next sections is 1) to investigate whether or not a specific range of ρ_s values provides a valuable gain in peak capacity for the same analysis time at constant pressure and at constant speed and 2) to validate the theoretical predictions from experimental data.

4.1.4. Peak capacity for the same analysis time

Before comparing the gradient performances of conical and cylindrical columns, it is important to choose the most appropriate metric for the resolution power of both types of columns. In this work, for the sake of comparison, two practical case scenario were considered: either the practitioner decides to operate at constant analysis time or at constant back pressure. In the first case scenario,

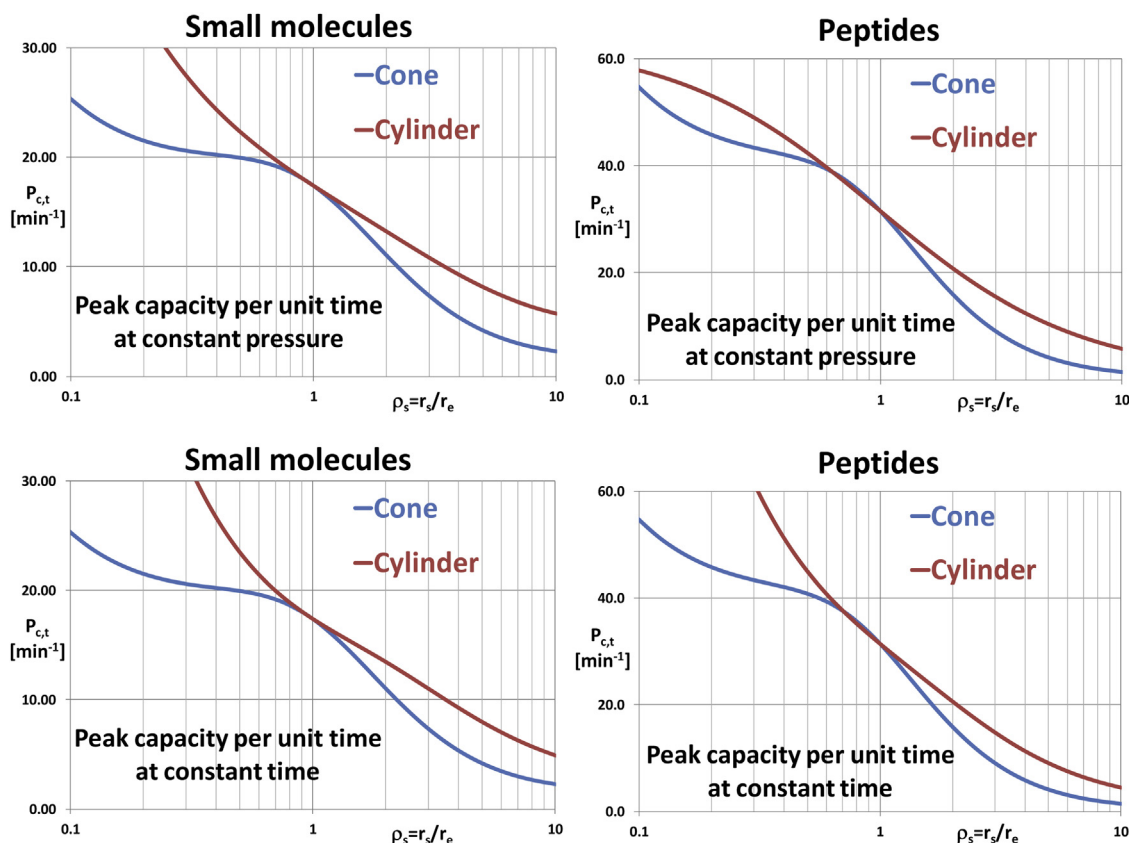


Fig. 2. Comparison between the gradient peak capacity for the same analysis time, $P_{c,t}$, of conical (solid blue lines) and cylindrical (solid red lines) columns packed with the same $5\ \mu\text{m}$ particles ($H=12\ \mu\text{m}$) and having the same length ($L=15\ \text{cm}$) as a function of the ratio, ρ_s , of the outlet column radius to the inlet column radius. The inlet radius of the conical column is fixed at $r_e=1.05\ \text{mm}$. The flow rate is set at $0.40\ \text{mL/min}$. Gradient time: $5\ \text{min}$. (Left) Small molecule mixture: $S=5$. Gradient steepness: $0.07\ \text{min}^{-1}$. (Right) Peptide mixture: $S=25$. Gradient steepness: $0.09\ \text{min}^{-1}$. (Top) Comparison at constant pressure. (Bottom) Comparison at constant time.

the absolute peak capacity or the peak capacity for the same analysis time are obviously equivalent. In the second case scenario, the analysis times are different for conical and cylindrical columns, so, the absolute peak capacities can no longer be compared. In contrast, the peak capacity for the same analysis time still provides a relevant metric to compare the resolution power of the two types of column, irrespective of the analysis time (which is inevitably longer for cones than it is for cylinders). Therefore, in this work, the peak capacity for the same analysis time was selected and calculated for both types of columns, irrespective of the case scenario.

In this section, the inner radius of the cylindrical column is either $r_{eq} = \rho_s r_e$ (see Eq. (10), comparison at constant pressure) or $r_{eq} = r_e \sqrt{\frac{1+\rho_s+\rho_s^2}{3}}$ (see Eq. (18), comparison at constant analysis time). The temporal peak widths observed at the column outlet (see Eq. (47)) result from the trade-off between two antagonist phenomena. The first one relates to the spatial compression ($\rho_s > \tilde{\rho}_s$) or expansion ($\rho_s < \tilde{\rho}_s$) of the chromatographic zone caused by the changes of retention and column i.d. across the peak width. The second one pertains to the linear velocity of the analyte at the column outlet which is directly proportional to $\frac{1}{\rho_s^2(1+k_e(\rho_s))}$. The question is then asked for which column hardware configuration or ρ_s value, if any, these two phenomena deliver a higher peak capacity for the same analysis time at the same pressure compared to that of a standard cylindrical column. The numerical calculation of the integrals in Eq. (46) will provide a quantitative and definitive answer to this question.

In this section, two models of sample mixtures are considered: the first one contains seventeen small molecules ($S=5$ and $H=12\ \mu\text{m}$) while the second is made of seventeen peptides ($S=25$

and $H=12\ \mu\text{m}$). The entrance i.d. of the conical column and the column length are fixed at $2.1\ \text{mm}$ and $15\ \text{cm}$, respectively. The flow rate is set at $0.40\ \text{mL/min}$, so, the linear velocity is $u_0(0)=17.77\ \text{cm/min}$ (for a total porosity assumed to be 65%). The gradient time is maintained at $5\ \text{min}$. The gradient amplitudes are 0.35 and 0.45 for the small molecule and peptide sample mixtures, respectively (see more details in the experimental section). Accordingly, $G=0.29$ (small molecules) and 1.90 (peptides) at the column inlet. The only variable left in the calculation of the peak capacity is ρ_s . For each value of ρ_s , a reference cylindrical column is unambiguously defined so that it generates either the same pressure resistance or the same analysis time as those of the conical column. Twenty-six $\rho_{s,j}$ ($1 < j < 26$) values are distributed in a range from 0.1 to 10 :

$$\rho_{s,j} = \frac{\rho_s}{100^{\frac{j-1}{25}}} \quad (63)$$

For each $\rho_{s,j}$ value, $k_{0,max,j}$ is unambiguously determined so that the elution time of the most retained analyte in the mixture for the conical column satisfies:

$$\tau_{e,max} = \tau_m + \tau_g \quad (64)$$

where τ_g is the dimensionless gradient time. For each compound $1 < i < 17$ in the sample mixture, $k_{0,i,j}$ is distributed as follows:

$$k_{0,i,j} = \frac{k_{0,max,j}}{100^{\frac{i-1}{16}}} \quad (65)$$

The peak capacities for the same analysis time, $P_{c,t}$, reported in Fig. 2 are calculated from Eq. (48) according to the predicted temporal standard deviations, $\sigma_{t,i,j}$, of each compound i given by Eq. (47) and from the elution times of the most retained ($k_{0,1,j}$) and

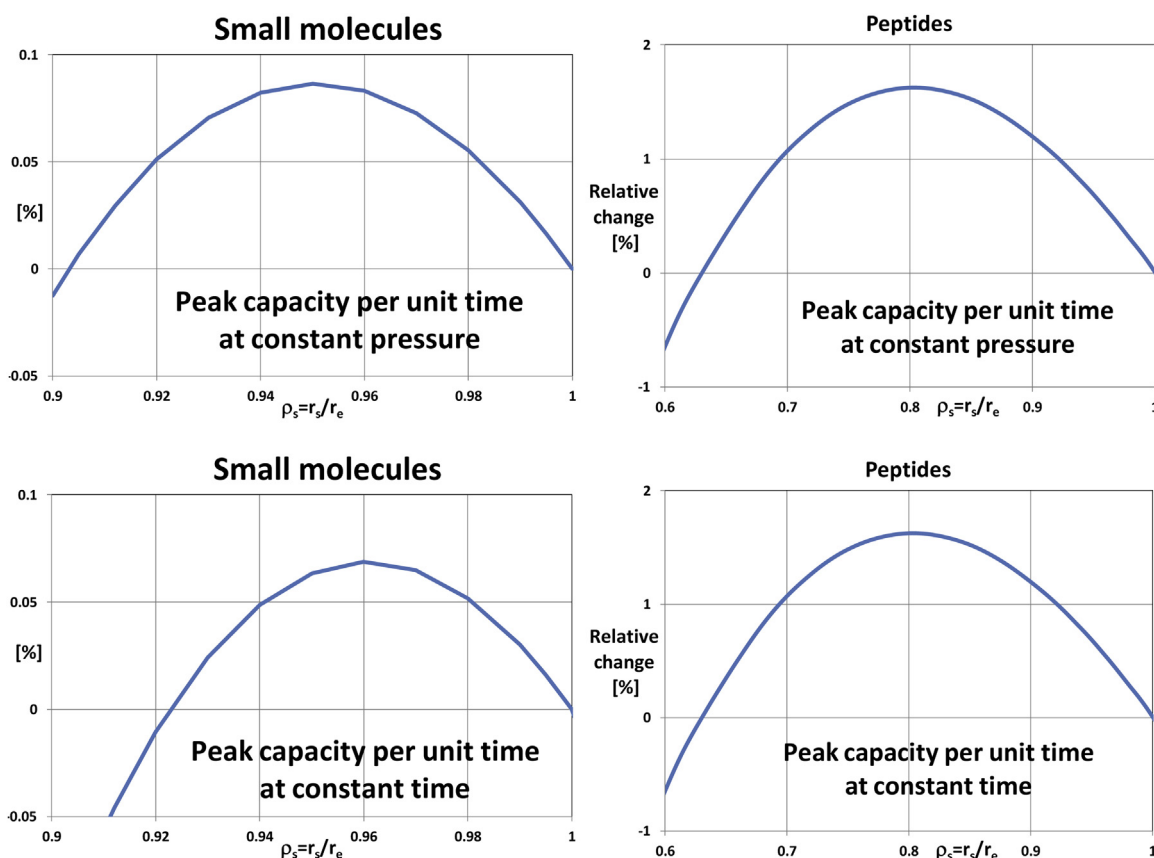


Fig. 3. Same plots as those in Fig. 2, except the expected relative gain in peak capacity for the same analysis time of the conical column relative to the performance of the cylindrical column is shown over a interval of ρ_s values for which conical columns slightly outperform the cylindrical column.

least retained ($k_{0.17j}$) compounds in the sample mixture. The results are compared at constant pressure (left graphs) and at constant analysis time (right graphs).

Regardless of the sample mixture, the two curves intersect at $\rho_s = 1$ because they both represent the same cylindrical column. For small molecules, the calculations apparently predict that $P_{c,t}$ is always smaller for conical than for the reference cylindrical. For instance, at constant pressure, the relative losses in column performance are 16.4% and 10.6% for $\rho_s = 2.0$ and 0.5, respectively (small molecules). The same relative losses are 23.3% and only 3.5% for peptides. At constant analysis time, they are 18.0% and 15.1% for small molecules and 23.0% and 9.3% for peptides. It is noteworthy that the asymmetry of the plots of $P_{c,t}$ with respect to $\rho_s = 1$ is consistent with a value of $\tilde{\rho}_s \neq 1$ for which the bandwidth of the analytes is neither compressed nor enlarged. For ρ_s in the range from 0.5 to 2, $\tilde{\rho}_s = 0.91$ –0.93 and 0.78–0.88 for small molecules and peptides, respectively. This means that flow direction definitely matters in gradient elution when using conical columns. They should be preferentially operated by applying the flow direction from the wide end to the narrow end of the column. Interestingly, the calculations even reveal that the conically shaped column slightly outperform the cylindrical column (+1.5% maximum, $\rho_s = 0.80$ –0.81) for $0.6 < \rho_s < 1$ and peptide mixtures. For small molecule mixtures, a small relative gain (+0.1% maximum, $\rho_s = 0.95$ –0.96) is still predicted within a narrower range of $0.9 < \rho_s < 1$. Fig. 3 show the positive relative gains of peak capacity for the same analysis time expected over these ρ_s intervals. It is noteworthy that $\tilde{\rho}_s$ provides a good estimate of the optimum ρ_s value for maximum relative gain in peak capacity for the same analysis time. Fig. 4 shows example of gradient chromatograms expected for conical ($\rho_s = 2$ and 0.5) and conventional columns operated at either constant pressure or constant analysis time.

Next, these predictions are tested experimentally. A stainless steel conical tube was prepared in-house, packed with 5 μm particles, and tested in isocratic and gradient conditions. ρ_s was fixed at 2 (or at 0.5 when reversing flow direction) and $L = 15$ cm. The two i.d. measured at both ends of this 15 cm long conical column are 2.1 mm and 4.2 mm (the opening angle of the truncated cone is then 0.4°).

4.2. Observations

4.2.1. Apparent plate heights of non-uniform columns at constant and variable local plate heights

The determination of the plot of the local plate height, $H(\zeta)$, vs. the dimensionless axial position, ζ , along the conical column operated at a fixed flow rate F_v is based on the assumption that the conical column (inner radii r_e and r_s) is as well packed as a reference cylindrical column of inner radius r_c . The dependence of the plate height on the superficial linear velocity is directly measured on the reference column. The intensity of the superficial linear velocity along the conical column is written:

$$u_{s,\text{cone}}(\zeta) = \frac{F_v}{\pi r_e^2 (1 + [\rho_s - 1]\zeta)^2} \quad (66)$$

Therefore, the actual variation of $u_{s,\text{cone}}(\zeta)$ along the conical column is bound by the extreme superficial linear velocities at the column inlet, $\frac{F_v}{\pi r_e^2}$, and at the column outlet, $\frac{F_v}{\pi r_s^2}$. The very same linear velocity $u_{s,\text{cone}}(\zeta)$ can be applied to the reference cylindrical column when a flow rate, $F_{v,c}(\zeta)$, applied to the cylindrical column obeys the following relationship:

$$\frac{F_{v,c}(\zeta)}{\pi r_c^2} = \frac{F_v}{\pi r_e^2 (1 + [\rho_s - 1]\zeta)^2} \quad (67)$$

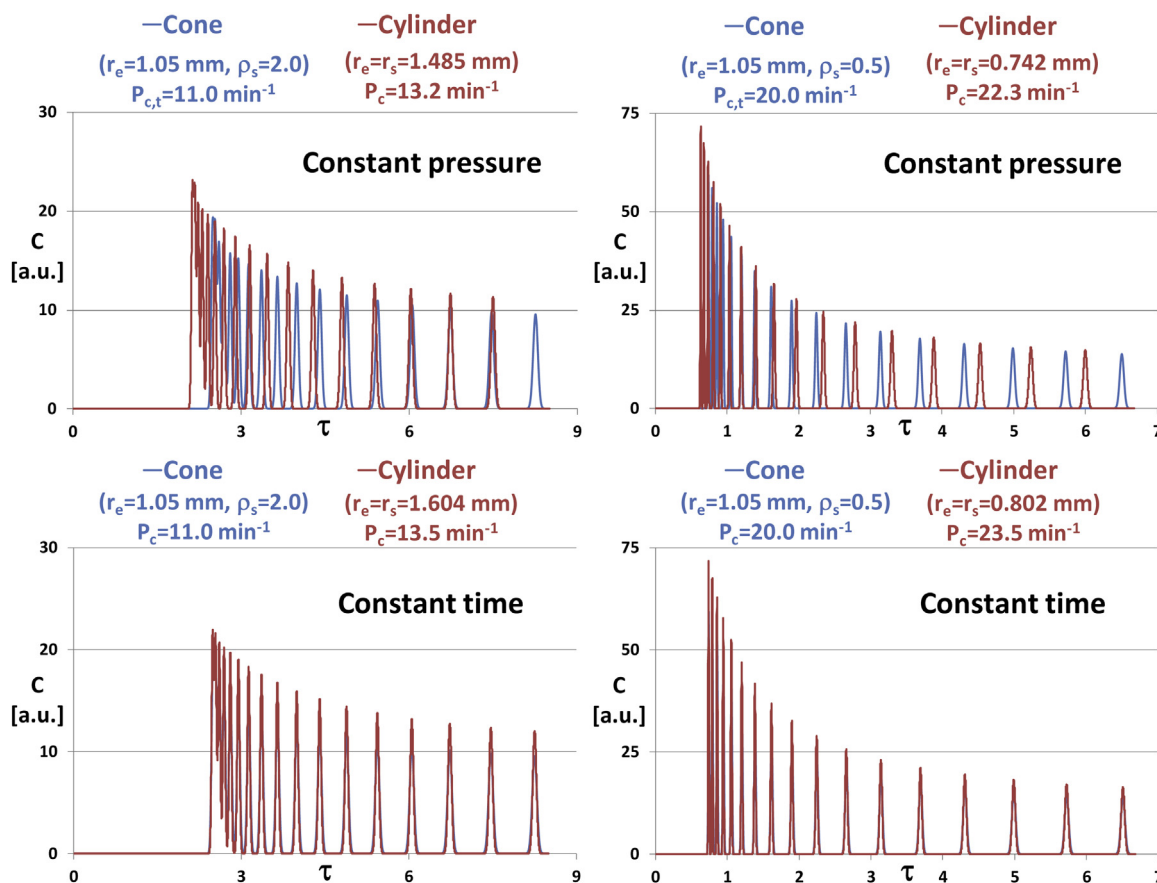


Fig. 4. Calculated gradient chromatograms of 17 small molecules ($S=5$) on both conical (solid blue line) and classical cylindrical (solid brown line) columns having the same local plate height H (12 μm), the same length L (15 cm), and operated at the same flow rate (0.4 mL/min). The entrance inner diameter of the conical column is $r_e = 1.05$ mm. Gradient time: 5 min. The temporal gradient steepness is $\beta = 0.07 \text{ min}^{-1}$. (Left graphs) $\rho_s = 2$. (Right graphs) $\rho_s = 0.5$. (Top graphs) Comparison at constant pressure. (Bottom graphs) Comparison at constant analysis time.

Accordingly, the variation of the experimental local plate height, $H(\zeta)$, of the five n -alkanophenones as a function of the dimensionless axial distance ζ along the 4.2 mm i.d. inlet / 2.1 mm i.d. outlet $\times$ 150 mm conical column when operated at the flow rate of $F_v = 0.40$ mL/min is represented in Fig. 5. The reference cylindrical column is the 3.0 mm $\times$ 150 mm cylindrical column. Based on Eq. (67), $H(0)$, $H(0.193)$, $H(0.367)$, $H(0.586)$, $H(0.735)$, $H(0.845)$, $H(0.931)$, and $H(1)$ were obtained by measuring the plate height of the five n -alkanophenones at seven different flow rates increasing from $F_v = 0.204$ to 0.306, 0.408, 0.510, 0.612, 0.714, and to 0.816 mL/min, respectively. The solid cyan line is the best curve describing the variation of the plate height of valerophenone (open cyan squares) along the conical column. For the sake of comparison, the plate height of valerophenone ($H = 9.50 \mu\text{m}$) measured on the cylindrical column at 0.35 mL/min (applied flow rate to generate the same analysis times as those of the conical column) is represented by the thick, dotted, cyan, and horizontal line. The symmetric graph with respect to $\zeta = 0.5$ is expected when reversing the flow direction (2.1 mm i.d. inlet/4.2 mm i.d. outlet $\times$ 150 mm conical column).

In order to compare the overall performance of both columns at constant analysis time, the applied flow rates were adjusted in order to compensate for the difference between the tube volume of the conical column (1.21 cm^3 , 0.40 mL/min) and that of the cylindrical column (1.06 cm^3 , 0.35 mL/min). In so doing, very similar retention times are observed on both columns. The column-to-column comparison is discussed in depth for the analyte valerophenone. Inevitably, the system pressure is expected to be slightly higher for the conical column (1190 psi) than that for the conventional column (1080 psi). The apparent plate height, $\bar{H}$,

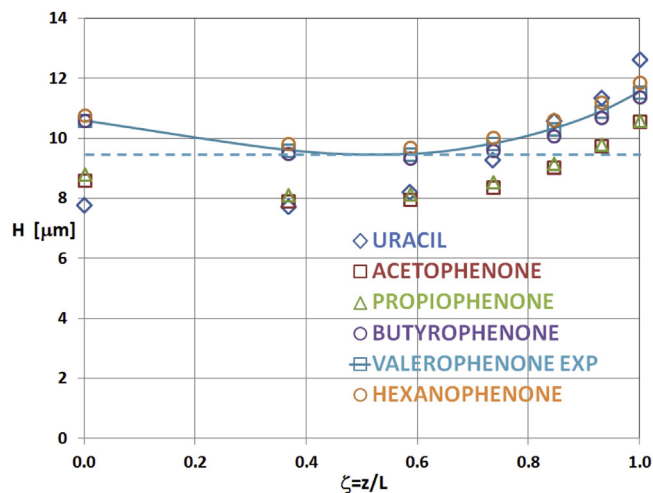


Fig. 5. Variation of the local plate height, $H(\zeta)$, of five alkanophenones as a function of the dimensionless axial position $\zeta = \frac{z}{L}$ along the conical column ($r_e = 2.1$ mm, $\rho_s = 0.5$, $L = 15$ cm) packed with 5 μm XBridge BEH-C₁₈ particles and run at a flow rate of 0.40 mL/min. The mobile phase is a mixture of acetonitrile and water (75/25, v/v) and the column temperature was set at 27 °C. The solid purple line is the best curve describing the variation of the plate height of valerophenone (open cyan squares) along the conical column. For the sake of comparison, the plate height of valerophenone ($H = 9.50 \mu\text{m}$) measured on the 3.0 mm i.d. cylindrical column at 0.35 mL/min is represented by the thick, dotted, cyan, and horizontal line.

of the conical column at 0.40 mL/min is estimated from Giddings' theory of non-uniform columns [2]. It is expected from Fig. 5 (varia-

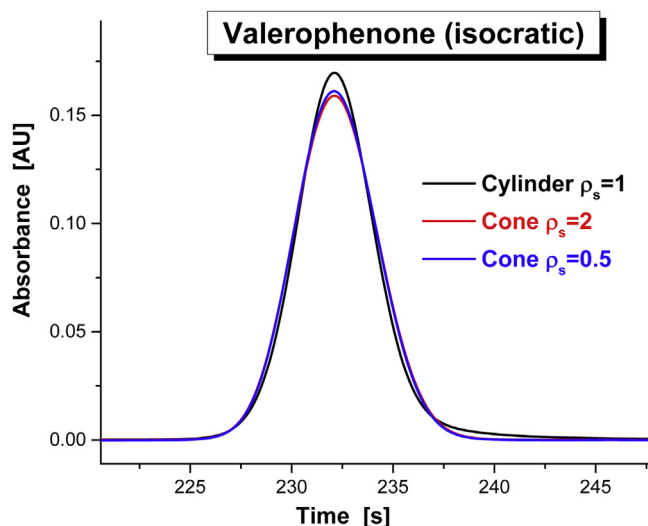


Fig. 6. Experimental peak profiles of valerophenone on the cylindrical column ($r_e = 1.50$ mm, $\rho_s = 1$) and on the conical column used in both flow directions ($r_e = 1.05$ mm and $\rho_s = 2$; $r_e = 2.1$ mm and $\rho_s = 0.5$) under isocratic condition. See detailed experimental conditions in the text. Note the smaller half-height peak width but the larger tailing on the conventional than on the conical column. See Table 1 for the resulting efficiencies calculated from the half-height peak width and from the first and second central moments of the concentration distributions.

tion of the local plate height along the conical column) and Eq. (20) (accounting for the variations of both the local plate height and the linear velocity) that $\bar{H} = 11.6$ μm provided that the conical column is as well packed as the cylindrical column. The plate height of the cylindrical column at 0.35 mL/min is measured at $H = 9.5$ μm . This means that under isocratic conditions, the efficiency of the conical column is 18.1% smaller than that of the cylindrical column. If the plate height was flow rate independent and equal to 9.5 μm , then, $\bar{H} = 10.8$ μm according to Eq. (25) and the efficiency of the conical column would be only 12.1% smaller than that of the cylindrical column.

4.2.2. Isocratic elution

Fig. 6 shows the experimental peak profiles of *n*-valerophenone recorded on the cylindrical (0.35 mL/min) and conical column (0.40 mL/min flow rate) under isocratic condition. Note that, for the sake of comparison, the time position of the three peaks were slightly adjusted in order to better appreciate the differences in peak width and peak shape. All the experimental data (system pressure, retention time, zeroth moment, first moment μ_1 , second central moment μ'_2 , half-height peak width $w_{1/2}$, efficiency based on half height peak width $N_{1/2}$, and efficiency based on the peak moments $N_{moments}$) are gathered in Table 1 for the other four compounds (acetophenone, propiophenone, butyrophenone, and hexanophenone) present in the sample mixture. The same conical column was used in both flow directions ($\rho_s = 0.5$ and 2). The column efficiencies based on the half-height peak width ($N_{1/2}$) and on the peak moments ($N_{moments}$) and reported in Table 1 are defined by (duplicate data):

$$N_{1/2} = 5.545 \frac{t_R^2}{w_{1/2}^2} \quad (68)$$

$$N_{moment} = \frac{\mu_1^2}{\mu'_2} \quad (69)$$

Consistent with Giddings' theory of non-uniform columns, the column efficiencies measured from the half-height peak widths for the conical column ($\rho_s = 0.5$) are 21.0% (acetophenone, $k = 0.35$), 19.4% (propiophenone, $k = 0.54$), 16.7% (butyrophenone,

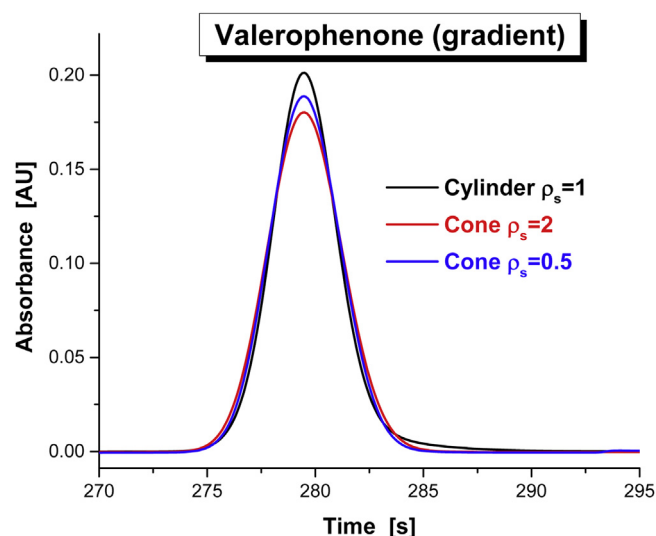


Fig. 7. Same as in Fig. 6 except the columns are run under gradient conditions. See Table 2 for the resulting peak capacities calculated from the half-height peak width and from the first and second central moments of the concentration distributions.

$k = 0.76$), 15.7% (valerophenone, $k = 1.08$), and 14.9% (hexanophenone, $k = 1.56$) lower for the 2.1/4.2 mm i.d. conical column than those measured for the cylindrical 3.0 mm i.d. column ($\rho_s = 0.5$). They are slightly larger (22.8%, 21.5%, 19.2%, 18.1%, and 17.2%, respectively) for the same conical column when reversing flow direction, e.g., for $\rho_s = 2$. In theory, they should have the same efficiency unless wall effects, radial component of the linear velocity vector, sample distribution at column inlet, and sample collection at column outlet are not equivalent in both flow directions. Overall, these observation clearly demonstrate that 1) the conical column is nearly as well packed as the cylindrical one (18.1% loss predicted) and 2) the observed relative decay in column efficiency is caused by both the conical shape of the stainless steel tube (e.g., the continuous change of flow velocity along the column, 12.1% efficiency decay if the plate is assumed to be uniform) and by the flow dependence of the local plate heights of these small molecules (causing the additional heights 15.7%–12.1% = 3.6% efficiency decay for valerophenone) as shown in Fig. 5. Indeed, according to Giddings' theory of non-uniform columns, both axial heterogeneities in linear velocity and plate height along the column negatively affect its apparent efficiency.

Remarkably, the peaks recorded on the standard cylindrical column systematically tails more than those observed for the conical column, irrespective of the flow direction (see Fig. 6): the column efficiencies measured from the peak moment analysis are 67% (acetophenone), 51% (propiophenone), 43% (butyrophenone), 41% (valerophenone), and 26% (hexanophenone) higher for the 2.1/4.2 mm i.d. conical than for the cylindrical column. Finally, regarding the impact of flow direction along the same conical column, it is noteworthy that the efficiencies are always slightly higher (+2.5%) for $\rho_s = 0.5$ (e.g., flow from the wide inlet i.d. to the narrow outlet i.d. of the conical column) than those for $\rho_s = 2$. In other words, the isocratic performance of packed conical columns is not fully independent on the flow direction as predicted by the theory. This non-ideality could be explained by the fact that wall and border effects in the conical column are not equivalent in both flow directions.

4.2.3. Gradient elution

Fig. 7 shows the experimental peak profiles of *n*-valerophenone recorded on the cylindrical (0.35 mL/min flow rate) and conical column (0.40 mL/min) under gradient condition. All the experi-

Table 1

Average retention times, zeroth moment, first moment, second central moment, half-height peak width, efficiency based on half height peak width ($N_{1/2}$), and efficiency based on the peak moments (N_{moments}) for five small molecules (alkanophenones) measured under isocratic (see experimental section for more details on the chromatographic conditions). Two duplicate set of experiments. Three column geometries are considered: 1) standard cylindrical column ($\rho_s = 1$), 2) conical column ($\rho_s = 2$), and 3) conical column ($\rho_s = 0.5$) after reversing the flow direction.

Cylindrical 3.0 mm × 150 mm column ($\rho_s = 1$) - $P_{\text{system}} = 1080$ psi							
Compound	Retention Time [min]	Zeroth Moment [a.u.]	First Moment [min]	Second Moment [min ²]	Width @ 50% [min]	$N_{1/2}$	N_{moments}
Acetophenone	2.491	1038000	2.495	0.00071	0.0422	19359	8822
Propriophenone	2.843	797500	2.847	0.00083	0.0486	18975	9723
Butyrophenone	3.265	885500	3.271	0.00134	0.0601	16387	8012
Valerophenone	3.865	811000	3.869	0.00156	0.0718	16090	9596
Hexanophenone	4.758	811000	4.762	0.00214	0.0896	15633	10594
Conical 2.1 → 4.2 mm × 150 mm column ($\rho_s = 2$) - $P_{\text{system}} = 1190$ psi							
Compound	Retention Time [min]	Zeroth Moment [min.mV]	First Moment [min]	Second Moment [min ²]	Width @ 50% [min]	$N_{1/2}$	N_{moments}
Acetophenone	2.520	1035000	2.522	0.00044	0.0486	14939	14532
Propriophenone	2.874	796500	2.875	0.00057	0.0555	14891	14476
Butyrophenone	3.297	884500	3.299	0.00096	0.0675	13245	11337
Valerophenone	3.898	812000	3.900	0.00114	0.0800	13181	13342
Hexanophenone	4.793	810000	4.795	0.00174	0.0992	12945	13214
Conical 4.2 → 2.1 mm × 150 mm column ($\rho_s = 0.5$) - $P_{\text{system}} = 1190$ psi							
Compound	Retention Time [min]	Zeroth Moment [min.mV]	First Moment [min]	Second Moment [min ²]	Width @ 50% [min]	$N_{1/2}$	N_{moments}
Acetophenone	2.528	1035000	2.529	0.00043	0.0482	15285	14874
Propriophenone	2.883	796500	2.884	0.00056	0.0549	15291	14879
Butyrophenone	3.309	883500	3.311	0.00094	0.0667	13643	11613
Valerophenone	3.915	811500	3.917	0.00113	0.0792	13563	13635
Hexanophenone	4.816	809000	4.818	0.00172	0.0983	13307	13535

Table 2

Average retention times, zeroth moment, first moment, second central moment, half-height peak width, peak capacity for the same analysis time ($P_{\text{c,t},1/2}$) based on half height peak width, and peak capacity for the same analysis time ($P_{\text{c,t},\text{moments}}$) based on the peak moments measured for five small molecules (alkanophenones) under gradient conditions (see experimental section for more details on the chromatographic conditions). Two duplicate set of experiments. Three column geometries are considered: 1) standard cylindrical column ($\rho_s = 1$), 2) conical column ($\rho_s = 2$), and 3) conical column ($\rho_s = 0.5$) after reversing the flow direction.

Cylindrical 3.0 mm × 150 mm column ($\rho_s = 1$) - $P_{\text{system}} = 1390 - 850$ psi							
Compound	Retention Time [min]	Zeroth Moment [a.u.]	First Moment [min]	Second Moment [min ²]	Width @ 50% [min]	$P_{\text{c,t},1/2}$ [min ⁻¹]	$P_{\text{c,t},\text{moments}}$ [min ⁻¹]
Acetophenone	2.927	1290000	2.930	0.00062	0.0433	28.4	21.4
Propriophenone	3.470	952500	3.473	0.00079	0.0478		
Butyrophenone	4.024	1020000	4.029	0.00118	0.0543		
Valerophenone	4.656	897500	4.660	0.00096	0.0570		
Hexanophenone	5.352	846000	5.356	0.00092	0.0586		
Conical 2.1 → 4.2 mm × 150 mm column ($\rho_s = 2$) - $P_{\text{system}} = 1540 - 940$ psi							
Compound	Retention Time [min]	Zeroth Moment [min.mV]	First Moment [min]	Second Moment [min ²]	Width @ 50% [min]	$P_{\text{c,t},1/2}$ [min ⁻¹]	$P_{\text{c,t},\text{moments}}$ [min ⁻¹]
Acetophenone	2.945	1105000	2.946	0.00049	0.0517	24.0	23.9
Propriophenone	3.483	815500	3.484	0.00060	0.0568		
Butyrophenone	4.032	862500	4.034	0.00078	0.0639		
Valerophenone	4.659	763000	4.661	0.00080	0.0669		
Hexanophenone	5.351	720000	5.352	0.00083	0.0686		
Conical 4.2 → 2.1 mm × 150 mm column ($\rho_s = 0.5$) - $P_{\text{system}} = 1540 - 940$ psi							
Compound	Retention Time [min]	Zeroth Moment [min.mV]	First Moment [min]	Second Moment [min ²]	Width @ 50% [min]	$P_{\text{c,t},1/2}$ [min ⁻¹]	$P_{\text{c,t},\text{moments}}$ [min ⁻¹]
Acetophenone	2.952	1110000	2.954	0.00045	0.0494	25.3	25.4
Propriophenone	3.493	818500	3.494	0.00054	0.0540		
Butyrophenone	4.044	853500	4.046	0.00064	0.0605		
Valerophenone	4.674	762000	4.675	0.00072	0.0634		
Hexanophenone	5.367	718500	5.369	0.00074	0.0650		

mental data (system pressure, retention time, zeroth moment, first moment, second central moment, half-height peak width $w_{1/2}$, peak capacity based on half height peak width $P_{\text{c},1/2}$, and peak capacity based on the peak moments, $P_{\text{c},\text{moment}}$, are listed in Table 2. The 4σ peak capacities, $P_{\text{c},1/2}$ and $P_{\text{c},\text{moment}}$, are calculated as follows:

$$P_{\text{c},1/2} = 1 + 5 \times \sqrt{5.545} \frac{t_{R,5} - t_{R,1}}{4 \sum_{i=1}^5 w_{i,1/2}} \quad (70)$$

$$P_{\text{c},\text{moments}} = 1 + 5 \frac{\mu_{1,5} - \mu_{1,1}}{4 \sum_{i=1}^5 \sqrt{\mu'_{2,i}}} \quad (71)$$

The observed relative diminutions of the peak capacity for the same analysis time are 15.5% for $\rho_s = 2$ and only 10.1% for $\rho_s = 0.5$. The relative loss is remarkably much less pronounced for $\rho_s = 0.5$ than it is for $\rho_s = 2$. This contrasts with the average relative diminutions of the column efficiency of 19.8% ($\rho_s = 2$) and 17.6% ($\rho_s = 0.5$) for all five small alkanophenones. This demonstrates experimentally that performing gradients and selecting the flow direction

Table 3

Average retention times, zeroth moment, first moment, second central moment, and half-height peak width of the peptide bombesin measured under gradient conditions (see experimental section for more details on the chromatographic conditions). Two duplicate set of experiments. Three column geometries are considered: 1) standard cylindrical column ($\rho_s = 1$), 2) conical column ($\rho_s = 2$), and 3) conical column ($\rho_s = 0.5$) after reversing the flow direction.

Column configuration ρ_s	Retention Time [min]	Zeroth Moment [min.mV]	First Moment [min]	Second Moment [min ²]	Width @ 50% [min]	P _{system} [psi]
1 (cylinder)	4.798	46800	4.798	0.000288	0.0386	1700–1010
2 (cone 2.1 → 4.2)	4.786	46550	4.786	0.000363	0.0450	1890–1590
0.5 (cone 4.2 → 2.1)	4.797	46850	4.798	0.000313	0.0422	1890–1600

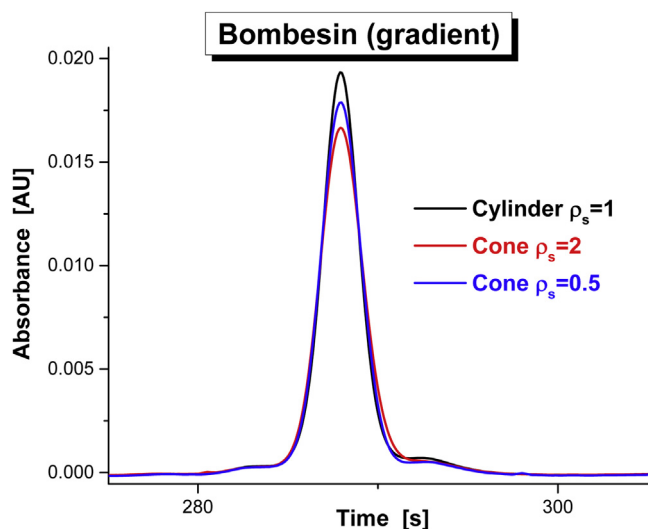


Fig. 8. Same as in Fig. 7 except the injected sample is the peptide bombesin. See the new gradient conditions in the text.

from the wide inlet to the narrow outlet of the conical column is beneficial in terms of gradient resolution with respect to the other flow direction. Gradient performance using conical columns depends then on the flow direction irrespective of wall and border effects. The experimental results are in qualitative agreement with the theoretical predictions when the actual flow rates (0.35 and 0.40 mL/min), column dimensions (3 mm i.d. and 2.1 mm/4.2 mm i.d. or 4.2 mm/2.1 mm), and change in plate height along the conical column (Fig. 5) are taken into account: the calculated relative losses in peak capacity for the same analysis time are 12.8% ($\rho_s = 2$) and only 5.4% ($\rho_s = 0.5$). The performance of the conical column under gradient elution is higher when imposing the flow rate from the wide inlet i.d. to the narrow outlet i.d.

Finally, Fig. 8 shows the experimental peaks of the peptide bombesin recorded on both columns under gradient conditions. The system pressure, retention time, zeroth moment, first moment, second central moment, and half-height peak width $w_{1/2}$ of bombesin are listed in Table 3 for the 3 column configurations. It confirms the results observed for the small molecules: the conical column performs better when the flow direction is from the wide to the narrow end of the column (relative increase of the peak width of 16.6% for $\rho = 2$ and only 9.3% for $\rho_s = 0.5$ with respect to the cylindrical column). If the theoretical predictions account for the actual flow rates (0.35 and 0.40 mL/min), column dimensions (3 mm i.d. and 2.1 mm/4.2 mm i.d. or 4.2 mm/2.1 mm), and change in plate height along the conical column (Fig. 5), it actually predicts a reduction of peak capacity for the same analysis time of 18.2% for $\rho_s = 2.0$ and even an increase of 2.9% for $\rho_s = 0.5$.

5. Conclusion

In this work, the potential of conically shaped analytical columns was investigated in both isocratic and gradient elution modes. The theoretical impact of the column geometry on speed, effi-

ciency, and gradient peak capacity was based on Giddings' (isocratic conditions) and Blumberg's (gradient conditions) theoretical developments of band broadening along non-uniform columns. The relative performance of the conical column were analyzed at either constant pressure or constant analysis time with respect to that of a cylindrical column, which has the same length and is packed with the same packing material.

As expected, both theory and experiments confirmed that, under isocratic conditions, the performance of the conical column cannot match that of standard cylindrical columns. Yet, the relative losses in speed and efficiency are smaller than 15% if the i.d. of the two column ends differ by less than a factor two. Interestingly, the gradient theory elaborated in this work predicts that conical columns could be slightly advantageous over conventional cylindrical columns if their outlet i.d. is smaller than their inlet i.d. The optimum ratio of the outlet to inlet i.d. is 0.95 and 0.80 for small molecule and peptide mixtures, respectively. Yet, the maximum relative gain does not exceed a few percents. Theory and experiments reveal that, in gradient conditions, the flow direction along a conical column should be from the wide end to the narrow end in order to generate the highest gradient performance.

Interestingly, the fabrication of the prototype conically shaped column revealed that conical columns reduce the peak tailing observed on conventional columns. Similar to the previously reported advantages of conical columns in preparative chromatography, conical columns can also be meaningful in analytical scale chromatography when it comes to minimize the nefarious impact of non-idealities (column border/wall effects) on column performance for cylindrical tubes. Given the expected reduction of the sample dispersion through the outlet frit, very short conical columns could be advantageously designed for ultra-fast gradient separations.

Appendix A. Supplementary Data

Supplementary data associated with this article can be found, in the online version, at <https://doi.org/10.1016/j.chroma.2019.01.055>.

References

- [1] J. Giddings, *Dynamics of Chromatography*, Marcel Dekker, New York, NY, 1965.
- [2] J.C. Giddings, Plate height of nonuniform chromatographic columns. gas compression effects, coupled columns, and analogous systems, *Anal. Chem.* 35 (1963) 353–356.
- [3] C. Ongkudon, S. Pan, M. Danquah, An innovative monolithic column preparation for the isolation of 25 kilo base pairs dna, *J. Chromatogr. A* 1318 (2013) 156–162.
- [4] C. Ongkudon, M. Danquah, Anion exchange chromatography of 4.2 kbp plasmid based vaccine (pcdna3f) from alkaline lysed *E. coli* lysate using amino functionalised polymethacrylate conical monolith, *Sep. Pur. Technol.* 78 (2011) 303–310.
- [5] W. Guan, F. Xu, F. Tan, Y. Guan, Study of conical columns with 10 degree opening angle for preparative liquid chromatography, *J. Chromatogr. A* 1108 (2006) 218–224.
- [6] J. Ma, L. Chen, Y. Guan, Study on conical columns with different conical angles for semi preparative liquid chromatography, *J. Chromatogr. A* 1033 (2004) 275–281.
- [7] M. Jiping, G. Yafeng, C. Lingxin, Study on conical columns for semi-preparative liquid chromatography, *J. Chromatogr. A* 26 (2003) 307–312.

- [8] L. Snyder, High Performance Liquid Chromatography - Advances and Perspectives, Vol. 1, Elsevier, Amsterdam, 1980.
- [9] L. Snyder, J. Dolan, High Performance Gradient Elution - The Practical Application of the Linear-Solvent-Strength Model, Wiley, Hoboken, 2007.
- [10] P. Jandera, J. Churacek, Gradient Elution in Column Liquid Chromatography-Theory and Practice, Elsevier, Amsterdam, 1985.
- [11] H. Poppe, Some reections on speed and efficiency of modern chromatographic methods, J. Chromatogr. A 778 (1997) 3–21.
- [12] H. Poppe, J. Paanakker, J. Bronckhorst, Peak width in solvent-programmed chromatography: I. general description of peak broadening in solvent-programmed elution, J. Chromatogr. 204 (1981) 77–84.
- [13] L. Blumberg, Variance of a zone migrating in a non-uniform time-invariant linear medium, J. Chromatogr. A 596 (1992) 1–13.
- [14] L. Blumberg, Variance of a zone migrating in a linear medium ii. time-varying non-uniform medium, J. Chromatogr. A 637 (1993) 119–128.
- [15] U.D. Neue, Theory of peak capacity in gradient elution, J. Chromatogr. A 1079 (2005) 153–161.
- [16] U. Neue, H.-J. Kuss, Improved reversed-phase gradient retention modeling, J. Chromatogr. A 1217 (2010) 3794–3803.
- [17] P. Nikitas, A. Pappa-Louisi, Expressions of the fundamental equation of gradient elution and a numerical solution of these equations under any gradient profile, Anal. Chem. 77 (2005) 5670–5677.
- [18] F. Gritti, G. Guiochon, The bandwidth in gradient elution chromatography with a retained organic modifier, J. Chromatogr. A 1145 (2007) 67–82.
- [19] F. Gritti, G. Guiochon, The distortion of gradient profiles in reversed-phase liquid chromatography, J. Chromatogr. A 1340 (2014) 50–58.
- [20] F. Gritti, G. Guiochon, Separations by gradient elution: Why are steep gradient profiles distorted and what is their impact on resolution in reversed-phase liquid chromatography, J. Chromatogr. A 1344 (2014) 66–75.
- [21] F. Gritti, General theory of peak compression in liquid chromatography, J. Chromatogr. A 1433 (2016) 114–122.
- [22] A. Said, Efficiency of composite vs. conically shaped chromatographic columns, J. High Resolut. Chromatogr. Commun. 3 (1979) 63–67.
- [23] J.M.A. Pecavar, I. Vovk, M. Prosek, Conically shaped chromatographic column, Anal. Sci. 1999 (2004) 233–240.
- [24] A. Prudkovski, Theory of chromatography for columns with a variable configuration, J. Anal. Chem. 2000 (2000) 443–444.
- [25] C. Poole, The essence of chromatography, Elsevier, Amsterdam, 2003.